

DEEPAK GHIMIRE, Ph.D. ([Interactive CV](#))

Computer Vision & AI/ML Expert | Senior Engineer & Researcher | 10+ Years in Edge AI, Autonomous Systems, ADAS, Robotics | Real-Time Inference, Sensor Fusion & Model Optimization

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Experience: 15+ Years (10+ after PhD)

Senior Research Engineer

Korea Electronics Technology Institute (KETI)

Jul 2024 – Present

Jeonju, Korea

KETI leads R&D in electronics and IT, supports technology commercialization, and provides infrastructure for testing and analysis. At KETI, my responsibilities include:

- Leading **research and industry-focused projects** that leverage AI, computer vision, and sensor technologies across agriculture, robotics, surveillance, and related domains.

Key Ongoing Projects

- Developed a crop monitoring system for **Smart Vertical Farming**, integrating RGB and hyperspectral cameras with temperature sensors for early disease detection and real-time crop health monitoring.
- Developing a **Smart Assist Robot** for large vehicles such as telehandlers, enabling them to scan narrow or unknown roads, detect obstacles, and propose optimal navigation routes using **LiDAR and camera sensors**.
- Developing **autonomous tractor guidance for paddy fields**, focusing on straight-line driving analysis and error measurement using GPS, IMU, and ZED 2i camera sensors.

Research Professor

Soongsil University

Jan 2022 – Jun 2024

Seoul, Korea

As a Research Professor, I was involved in research, strategic planning, course instruction, project acquisition, and scholarly publication. My key contributions included:

- Developed **efficient deep learning solutions for edge computing**, focusing on one-shot structured pruning and dataset compression to optimize model performance and resource utilization.
- Taught Computer Vision and Deep Learning**; supervised graduate students on AI/ML research projects.

AI / Computer Vision Engineer (Freelance)

BeEmotion.ai (formerly NVISO.ai), Switzerland

Feb 2018 – Apr 2022

Remote via Upwork Inc.

BeEmotion.ai develops AI-driven computer vision solutions for emotion recognition and behavior analysis, applied in automotive, healthcare, and consumer electronics. As a freelance engineer, I was involved in:

- Benchmarking deep neural network (DNN) models**, including backbones, face detection, and facial emotion recognition, using state-of-the-art inference tools across edge computing platforms like Raspberry Pi and NVIDIA Jetson Nano/Xavier.
- Contributing to the development of a vision-based automotive **in-car Interior Monitoring System (IMS)** AI solution. *Demo:* [Vimeo](#).

Education

Ph.D. in Computer Science and Engineering

Jeonbuk National University

Mar 2011 – Aug 2014

Jeonju, Korea

- Thesis:** *Recognition of Facial Expressions Based on Salient Texture and Geometric Features.*
- [Degree Certificate](#) | [Transcript](#)

M.Sc. in Computer Science and Engineering

Jeonbuk National University

Mar 2009 – Feb 2011

Jeonju, Korea

- Thesis:** *A Robust Face Detection Method Based on Skin Color and Edges.*
- [Degree Certificate](#) | [Transcript](#)

Technical Skills

- Programming Languages**
 - C/C++ (High-performance systems)
 - Python (AI/ML, scripting)
 - Java (Application development)
 - MATLAB (Scientific computing)
- Libraries and Frameworks**
 - PyTorch** / **TensorFlow** (Deep learning)
 - TensorRT** / **OpenVINO** (AI model optimization)
 - NCNN** / **TVM** / **Tengine** / **Paddle-Lite** / **MNN** (Lightweight AI inference)
 - OpenCV** / **Dlib** (Computer vision and machine learning)
 - Open3D** / **MMDetection3D** / **OpenPCDet** (3D data processing and object detection)
 - ARMCL** (Embedded ML acceleration)
 - VLFeat** (Vision library for MATLAB/C)
- Development Tools**
 - Qt (GUI applications)
 - Visual Studio (C++ development)
 - ROS1/ROS2 (Robotics middleware)
- Edge Computing Platforms**
 - Raspberry Pi (Embedded systems)
 - NVIDIA Jetson Nano/Xavier/Orin (AI at the edge)
- Model Deployment and MLOps**
 - ONNX Runtime, Docker, FastAPI
 - Git/GitHub, TensorBoard, W&B

Postdoctoral Researcher

Korea Electronics Technology Institute (KETI)

Jul 2014 – Feb 2020 Seoul, Korea

At KETI, my work focused on:

- Providing vision-based solutions for various challenges faced by government and private sector clients, including demonstrating ideas for project development and funding acquisition.
- Developing and implementing vision-based ADAS algorithms, including driver state monitoring, forward collision warning, and lane departure warning, and deploying them into vehicle front-facing dash cam systems and Android phones. Demo: YouTube.
- Contributing to multiple research projects focused on pedestrian and vehicle detection for surveillance monitoring, bird's eye view-based 3D object detection (Demo: YouTube), stereo vision-based depth estimation, and algorithm optimization.

Graduate Research Assistant

Jeonbuk National University

Mar 2009 – Aug 2014 Jeonju, Korea

- Contributed to multiple projects during my M.Sc. and Ph.D. studies, focusing on vision-based technologies such as face detection and recognition, facial emotion analysis, driver state monitoring, cast shadow detection, color image enhancement, object detection, and tracking.

Example Projects

- **Illegal Crossing Detection at Traffic Intersections**
Developed a real-time traffic monitoring system to detect illegal crossings by tracking pedestrians and vehicles at intersections.
Publication: IWPR 2016, Demo: YouTube
- **Touchless User Interaction for Large Displays**
Designed an IR camera-based system to detect fingertip positions and map them to display coordinates, enabling intuitive touchless interaction.
Publication: Int. J. Contents 2010, Demo: YouTube

Visiting PhD Scholar

Simon Fraser University, School of Computing Science

Feb 2014 – Apr 2014 Burnaby, BC, Canada

- Worked on facial expression analysis at the Vision and Media Lab, focusing on geometric facial feature extraction and tracking techniques.

Teaching Faculty

Lumbini Engineering College, Pokhara University

Aug 2007 – Nov 2008 Butwal, Nepal

- Taught courses in Object-Oriented Programming in C++, Computer Graphics, Digital Signal Processing, and Image Processing, including practical classes involving coding projects and hands-on practices.
- Supervised final-year undergraduate student projects as part of their degree requirements.

Academic Services

- **Editorial Board**
 - Discover Artificial Intelligence, Springer Nature.
- **Guest Editor for Special Issues**
 - Scale Space and Variational Methods in Computer Vision, MDPI, Applied Sciences (2022–2023).
 - Artificial Intelligence and Computer Vision in Neurological Disorders, MDPI, Diagnostics (2025–2026).
- **Distinguished Peer Reviewer**
 - Conducted over 60 verified peer reviews for multiple journals.
- **Technical Program Committee Member**
 - IWPR, International Workshop on Pattern Recognition, Tokyo, Japan, 11–13 May 2016.
 - ICSIP, IEEE International Conference on Signal and Image Processing, Beijing, China, 13–15 Aug. 2016.
- **Book Technical Reviewer**
 - Learning OpenCV 5 Computer Vision with Python – Fourth Edition (Authors: Joseph Howse, Joe Minichino), Packt Publishing Ltd.

Other Achievements

- **Academic Scholarships**
 - Full scholarships for M.Sc. and Ph.D. studies under the Brain Korea 21 Plus (BK21+) initiative (2009–2014).
 - Tuition waiver for undergraduate program (2003–2007) awarded to the top-performing student.
- **Publication Citation Metrics**
 - Citation data as of Sept 14, 2026.

Platform	Articles	h-index	Citations
Google Scholar	53	20	2016
ResearchGate	54	19	1641
Semantic Scholar	51	17	1396
Scopus	41	18	1368
Web of Science	36	12	772

Languages

Nepali (Native)	■■■■■■■■■■
English (C1 - Advanced)	■■■■■■■■■
Korean (Fluent)	■■■■■■■
Hindi (Fluent)	■■■■■■■

Proficiency levels are self-assessed.

Selected Publications

ID	Year	Title	Authors	Venue	IF/Q
J1	2026	STV-FSAnet: Track-Level Spatio-Temporal Verification for Fire and Smoke Alarm Validation in Video Surveillance	Ghimire, Kim, Jo, Lee, Jeong, Kim	Sensors	4.0/Q2
J2	2026	Vision-Based Strawberry Ripeness Grading for Harvest Decision Support in Smart Greenhouses	Lee, Ghimire, Cho, Seo, Jeong, Kim	Agriculture	4.5/Q1
C1	2026	One-cycle Structured Pruning with Stability Driven Subnetwork Search	Ghimire, Kil, Jeong, Park, Kim	WACV 2026 (IEEE/CVF)	-
J3	2026	Clinical and Machine Learning-Based Validation of CDAI Against DAS28-CRP in Rheumatoid Arthritis: A Cross-sectional Study in Nepal	Santosh, Paudyal, Ghimire, Sharma, BK	Under Review	-
J4	2026	Precision Biomarker Identification in Gynecological Cancers Using Coexpression Networks and Attention-Based LSTM in Healthcare 4.0	Sarker, Ahammed, Hosen, Miah, Islam, Ghimire, Hwang, Hosen	Diagnostics	3.3/Q1
J5	2025	Temporal Trends and Machine Learning-Based Risk Prediction of Female Infertility: A Cross-Cohort Analysis Using NHANES Data (2015–2023)	Begum, Ghimire, Hosen	Diagnostics	3.3/Q1
J6	2025	FUSE-Net: Multi-Scale CNN for NIR Band Prediction from RGB Using GNDVI-Guided Green Channel Enhancement	Lee, Ghimire, Kim, Cho, Kim, Jeong	Sensors	3.5/Q2
J7	2025	Association of Polycystic Ovary Syndrome with Clinical, Physical, and Reproductive Factors: A Data-Driven Analysis	Begum, Hosen, Ghimire, Park	Diagnostics	3.3/Q1
J8	2025	Federated Explainable AI-Based Alzheimer's Disease Prediction With Multimodal Data	Jahan, Adib, Kaiser, Hosen, Ghimire, Park	IEEE Access	3.6/Q2
J9	2024	Exploring Deep Learning and Machine Learning Approaches for Brain Hemorrhage Detection	Ahmed, Esha, Rahman, Kaiser, Hosen, Ghimire, Park	IEEE Access	3.6/Q2
J10	2024	Exploring Machine Learning for Predicting Cerebral Stroke: A Study in Discovery	Mia, Khanam, Ovy, Ghimire, Park, Begum, Hosen	Electronics	2.6/Q2
J11	2023	Loss-Aware Automatic Selection of Structured Pruning Criteria for Deep Neural Network Acceleration	Ghimire, Lee, Kim	Image and Vision Computing	4.2/Q1
J12	2023	A Review on the Detection Techniques of Polycystic Ovary Syndrome Using Machine Learning	Ahmed, Rahman, Jahan, Kaiser, Hosen, Ghimire, Kim	IEEE Access	3.6/Q2
J13	2023	RNN-CNN Based Cancer Prediction Model for Gene Expression	Thakur, Batra, Malik, Ghimire, Kim, Hosen	IEEE Access	3.6/Q2
J14	2023	Healthcare Internet of Things (H-IoT): Current Trends, Future Prospects, Applications, Challenges, and Security Issues	Kumar, Kumar, Verma, Ghimire, Kim, Hosen	Electronics	2.6/Q2
J15	2022	Magnitude and Similarity Based Variable Rate Filter Pruning for Efficient Convolution Neural Networks	Ghimire, Kim	Applied Sciences	2.5/Q2
J16	2022	A Survey on Efficient Convolutional Neural Networks and Hardware Acceleration	Ghimire, Kil, Kim	Electronics	2.5/Q2
C2	2017	Facial Expression Recognition Using Local Region Specific Dense Optical Flow and LBP Features	Ghimire, Park, Kim	ACIVS 2017 (LNCS, Springer)	-
C3	2016	Object Tracking via Background Subtraction for Monitoring Illegal Activity in Crossroad	Ghimire, Jeong, Park, Lee	IAPR PR 2016 (SPIE)	-
J17	2016	Facial Expression Recognition Based on Local Region Specific Features and SVMs	Ghimire, Jeong, Lee, Park	Multimedia Tools and Applications	0.78/Q1 (SJR)
J18	2016	Recognition of Facial Expressions Based on Salient Geometric Features and Support Vector Machines	Ghimire, Lee, Li, Jeong	Multimedia Tools and Applications	0.78/Q1 (SJR)
C4	2015	Facial Expression Recognition Based on Region Specific Appearance and Geometric Features	Ghimire, Jeong, Yoon, Choi,	ICDIM 2015 (IEEE)	-
J19	2015	Online Sequential Extreme Learning Machine-based Co-training for Moving Cast Shadow Detection	Ghimire, Lee	Multimedia Tools and Applications	0.78/Q1 (SJR)
C5	2014	A Local Technique for Contrast Preserving Medical Image Enhancement	Pant, Ghimire, Park, Lee	SPIE Medical Imaging 2014	-
J20	2013	Geometric Feature-Based Facial Expression Recognition in Image Sequences Using Multi-Class AdaBoost and Support Vector Machines	Ghimire, Lee	Sensors	3.5/Q2
C6	2013	Automatic Facial Expression Recognition Based on Features Extracted from Tracking of Landmarks	Ghimire, Lee	ICGPI 2013 (SPIE)	-
C7	2012	A Lighting Insensitive Face Detection Method on Color Images	Ghimire, Lee	SCET 2012 (IEEE)	-
C8	2011	Nonlinear Transfer Function-Based Image Detail Preserving Dynamic Range Compression for Color Image Enhancement	Ghimire, Lee	PSIVT 2011 (LNCS, Springer)	-
J21	2011	Nonlinear Transfer Function-Based Local Approach for Color Image Enhancement	Ghimire, Lee	IEEE Trans. on Consumer Electronics	10.9/Q1
C9	2010	Color Image Enhancement in HSV Space Using Nonlinear Transfer Function and Neighborhood Dependent Approach with Preserving Details	Ghimire, Lee	PSIVT 2010 (IEEE)	-

Note: C = Conference Proceeding; J = Journal Publication.